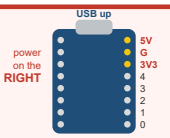
Top view, enlarged for reading

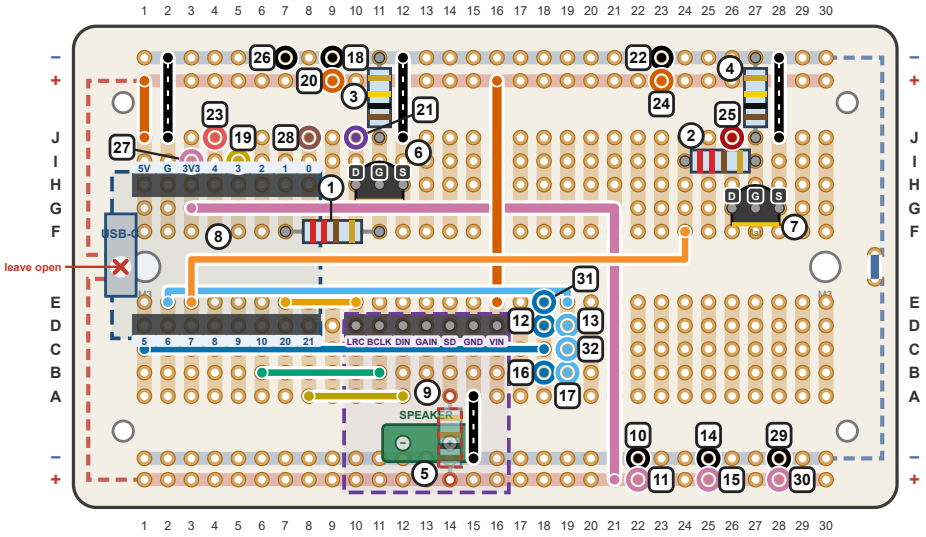
Intake panel, C3 POWER-RIGHT — protoboard

TOP VIEW (component side), enlarged, not to scale

For a C3 POWER-RIGHT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the RIGHT row. A power-left module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: F7, F11
2. RG2 220 Ω: I24, I27
3. RP1 100 kΩ: J11, top − rail at number 11
4. RP2 100 kΩ: J27, top − rail at number 27
5. RSD1 10 kΩ: bottom + rail at number 14, A14; fit only if the amplifier's SD pin reads near 0 V in the SD pull-up check
6. Q1 BS170: H10, H11, H12
7. Q2 BS170: G26, G27, G28
8. U1 ESP32-C3 Super Mini: D1-D8, H1-H8
9. U2 MAX98357 amplifier: D10-D16

Device wires

OLED

10. GND: bottom − rail at number 22
11. 3V3: bottom + rail at number 22
12. SDA: D18
13. SCL: D19

AIR

14. GND: bottom − rail at number 25
15. 3V3: bottom + rail at number 25
16. SDA: B18
17. SCL: B19

YELLOW

18. GND: top − rail at number 9
19. HOLD: I5
20. 5V: top + rail at number 9
21. LED− Y: J10

RED

22. GND: top − rail at number 23
23. DISC: J4
24. 5V: top + rail at number 23
25. LED− R: J26

BUZZER

26. GND: top − rail at number 7
27. 3V3: I3
28. BUZ: J8

NFC

29. GND: bottom − rail at number 28
30. 3V3: bottom + rail at number 28
31. SDA: E18
32. SCL: C19

Nets (wire colours)

- **GND** ground
- **SCL** I2C clock, GPIO 6
- **5V** USB 5 V (power)
- **BCLK** amplifier bit clock, GPIO 10
- **DIN** amplifier data, GPIO 21
- **HOLD** yellow button, GPIO 3
- **LAMP Y** yellow lamp drive, GPIO 1
- **GATE Y** Q1 gate
- **LED− Y** yellow lamp − / Q1 drain
- **SDA** I2C data, GPIO 5
- **BUZ** buzzer drive, GPIO 0
- **3V3** 3.3 V from the C3 (power)
- **LRC** amplifier word select, GPIO 20
- **SD** amplifier SD (on/off)
- **DISC** red button, GPIO 4
- **LAMP R** red lamp drive, GPIO 7
- **GATE R** Q2 gate
- **LED− R** red lamp − / Q2 drain

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- (9) a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves
- leave open: fitting a link here would join two rail halves that carry different nets (see the build steps)
- DNP: a part fitted only when needed (see the build list); keep its holes free
- a wire laid along the centre channel (no copper there), under U1 where drawn

This bar is NOT 20 mm on this enlarged page: measure on the real-size pages

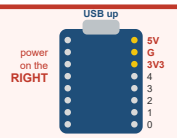
Solder side, enlarged for reading

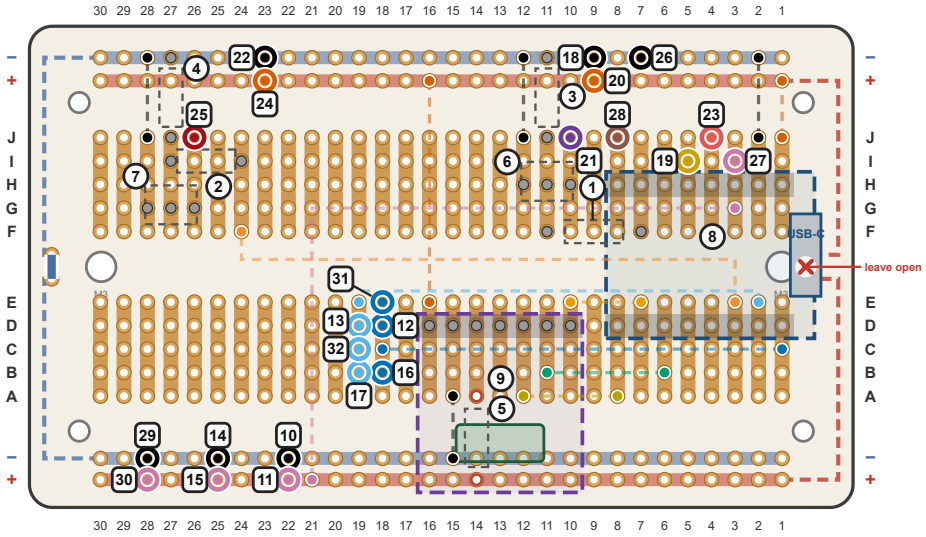
Intake panel, C3 POWER-RIGHT — protoboard

SOLDER SIDE (mirrored): number 1 is on the RIGHT

For a C3 POWER-RIGHT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the RIGHT row. A power-left module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





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AIR

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15. 3V3: bottom + rail at number 25
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YELLOW

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19. HOLD: I5
20. 5V: top + rail at number 9
21. LED− Y: J10

RED

22. GND: top − rail at number 23
23. DISC: J4
24. 5V: top + rail at number 23
25. LED− R: J26

BUZZER

26. GND: top − rail at number 7
27. 3V3: I3
28. BUZ: J8

NFC

29. GND: bottom − rail at number 28
30. 3V3: bottom + rail at number 28
31. SDA: E18
32. SCL: C19

Nets (wire colours)

- **GND** ground
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- **BCLK** amplifier bit clock, GPIO 10
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- **LAMP R** red lamp drive, GPIO 7
- **GATE R** Q2 gate
- **LED− R** red lamp − / Q2 drain

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- a wire laid along the centre channel (no copper there), under U1 where drawn

This bar is NOT 20 mm on this enlarged page: measure on the real-size pages

Real size (print at 100 %, "actual size"): lay the board on the drawing

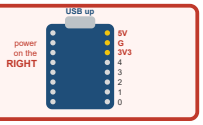
Check the 20 mm bar with a ruler before you rely on the drawing.

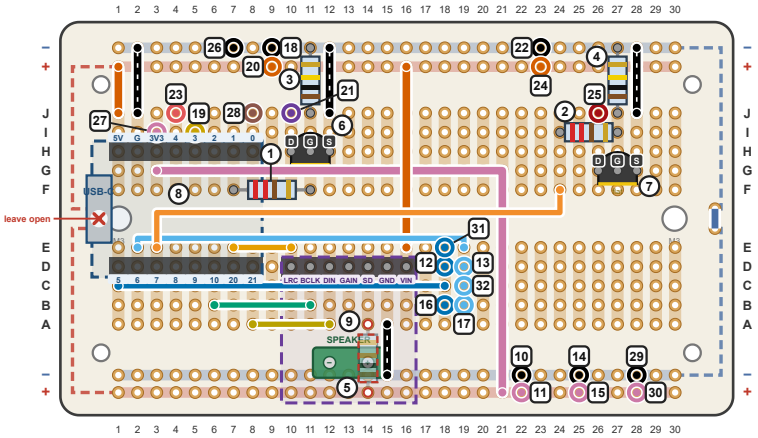
Intake panel, C3 POWER-RIGHT — protoboard

TOP VIEW (component side), real size

For a C3 POWER-RIGHT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the RIGHT row. A power-left module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: F7, F11
2. RG2 220 Ω: I24, I27
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5. RSD1 10 kΩ: bottom + rail at number 14, A14; fit only if the amplifier's SD pin reads near 0 V in the SD pull-up check
6. Q1 BS170: H10, H11, H12
7. Q2 BS170: G26, G27, G28
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11. 3V3: bottom + rail at number 22
12. SDA: D18
13. SCL: D19

AIR

14. GND: bottom − rail at number 25
15. 3V3: bottom + rail at number 25
16. SDA: B18
17. SCL: B19

YELLOW

18. GND: top − rail at number 9
19. HOLD: I5
20. 5V: top + rail at number 9
21. LED− Y: J10

RED

22. GND: top − rail at number 23
23. DISC: J4
24. 5V: top + rail at number 23
25. LED− R: J26

BUZZER

26. GND: top − rail at number 7
27. 3V3: I3
28. BUZ: J8

NFC

29. GND: bottom − rail at number 28
30. 3V3: bottom + rail at number 28
31. SDA: E18
32. SCL: C19

Nets (wire colours)

- **GND** ground
- **SCL** I2C clock, GPIO 6
- **5V** USB 5 V (power)
- **BCLK** amplifier bit clock, GPIO 10
- **DIN** amplifier data, GPIO 21
- **HOLD** yellow button, GPIO 3
- **LAMP Y** yellow lamp drive, GPIO 1
- **GATE Y** Q1 gate
- **LED− Y** yellow lamp − / Q1 drain
- **SDA** I2C data, GPIO 5
- **BUZ** buzzer drive, GPIO 0
- **3V3** 3.3 V from the C3 (power)
- **LRC** amplifier word select, GPIO 20
- **SD** amplifier SD (on/off)
- **DISC** red button, GPIO 4
- **LAMP R** red lamp drive, GPIO 7
- **GATE R** Q2 gate
- **LED− R** red lamp − / Q2 drain

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
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- a wire laid along the centre channel (no copper there), under U1 where drawn
- 20 mm: printed at 100 % (actual size) this bar measures 20 mm and the board is real size

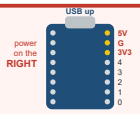
Real size, solder side

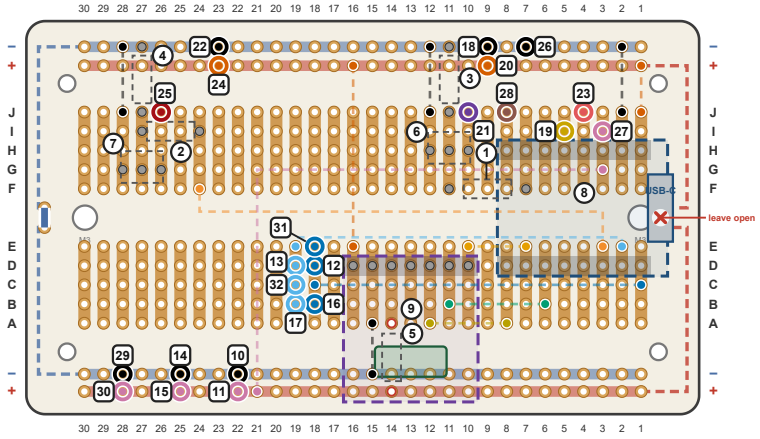
Intake panel, C3 POWER-RIGHT — protoboard

SOLDER SIDE (mirrored): number 1 is on the RIGHT

For a C3 POWER-RIGHT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the RIGHT row. A power-left module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: F7, F11
2. RG2 220 Ω: I24, I27
3. RP1 100 kΩ: J11, top − rail at number 11
4. RP2 100 kΩ: J27, top − rail at number 27
5. RSD1 10 kΩ: bottom + rail at number 14, A14; fit only if the amplifier's SD pin reads near 0 V in the SD pull-up check
6. Q1 BS170: H10, H11, H12
7. Q2 BS170: G26, G27, G28
8. U1 ESP32-C3 Super Mini: D1-D8, H1-H8
9. U2 MAX98357 amplifier: D10-D16

Device wires

OLED

10. GND: bottom − rail at number 22
11. 3V3: bottom + rail at number 22
12. SDA: D18
13. SCL: D19

AIR

14. GND: bottom − rail at number 25
15. 3V3: bottom + rail at number 25
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YELLOW

18. GND: top − rail at number 9
19. HOLD: I5
20. 5V: top + rail at number 9
21. LED− Y: J10

RED

22. GND: top − rail at number 23
23. DISC: J4
24. 5V: top + rail at number 23
25. LED− R: J26

BUZZER

26. GND: top − rail at number 7
27. 3V3: I3
28. BUZ: J8

NFC

29. GND: bottom − rail at number 28
30. 3V3: bottom + rail at number 28
31. SDA: E18
32. SCL: C19

Nets (wire colours)

- **GND** ground
- **SCL** I2C clock, GPIO 6
- **5V** USB 5 V (power)
- **BCLK** amplifier bit clock, GPIO 10
- **DIN** amplifier data, GPIO 21
- **HOLD** yellow button, GPIO 3
- **LAMP Y** yellow lamp drive, GPIO 1
- **GATE Y** Q1 gate
- **LED− Y** yellow lamp − / Q1 drain
- **SDA** I2C data, GPIO 5
- **BUZ** buzzer drive, GPIO 0
- **3V3** 3.3 V from the C3 (power)
- **LRC** amplifier word select, GPIO 20
- **SD** amplifier SD (on/off)
- **DISC** red button, GPIO 4
- **LAMP R** red lamp drive, GPIO 7
- **GATE R** Q2 gate
- **LED− R** red lamp − / Q2 drain

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- (9) a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves
- leave open: fitting a link here would join two rail halves that carry different nets (see the build steps)
- DNP: a part fitted only when needed (see the build list); keep its holes free
- a wire laid along the centre channel (no copper there), under U1 where drawn

20 mm: printed at 100 % (actual size) this bar measures 20 mm and the board is real size

Build order and hole table

For a C3 POWER-RIGHT module only (USB up: 5V, G, 3V3 at the top of the RIGHT row). A power-left module in this layout puts 5 V on a GPIO pin and can destroy it.

Build order (lowest parts first). The numbers are the callouts on the drawing; wire links carry no number:

- Rail link (−): wire from the right link pad, upper to the right link pad, lower. It joins the top and bottom − rails.
- Wire 3V3: G3 to bottom + rail at number 21 (one bend, 75 mm).
- Wire LAMP R: E3 to F24 (two bends, along the centre channel, 61 mm).
- Wire SCL: E2 to E19 (two bends, along the centre channel, 46 mm).
- Wire SDA: C1 to C18 (straight, 43 mm).
- Wire BCLK: B6 to B11 (straight, 13 mm).
- Wire DIN: A8 to A12 (straight, 10 mm).
- Wire LRC: E7 to E10 (straight, 8 mm).
- Wire GND: A15 to bottom − rail at number 15 (straight, 7 mm).
- Wire 5V: top + rail at number 16 to E16 (straight, 24 mm).
- Wire GND: J2 to top − rail at number 2 (straight, 9 mm).
- Wire GND: J12 to top − rail at number 12 (straight, 9 mm).
- Wire GND: J28 to top − rail at number 28 (straight, 9 mm).
- Wire 5V: J1 to top + rail at number 1 (straight, 6 mm).
- Part 1, RG1 (220 Ω): F7 to F11, lying flat.
- Part 2, RG2 (220 Ω): I24 to I27, lying flat.
- Part 3, RP1 (100 kΩ): J11 to top − rail at number 11, lying flat.
- Part 4, RP2 (100 kΩ): J27 to top − rail at number 27, lying flat.
- Part 6, Q1 (BS170): D in H10, G in H11, S in H12, flat face toward the bottom edge (the A side).
- Part 7, Q2 (BS170): D in G26, G in G27, S in G28, flat face toward the bottom edge (the A side).
- Part 8, U1 sockets: two 1x8 female headers in D1-D8, H1-H8. Plug U1 in while soldering to keep them square, then take it out.
- Part 9, U2 socket: a 1x7 female header in D10-D16. Plug U2 in while soldering to keep it square, then take it out.
- Device wires 10-13, OLED: 10 GND in bottom − rail at number 22, 11 3V3 in bottom + rail at number 22, 12 SDA in D18, 13 SCL in D19. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 14-17, AIR: 14 GND in bottom − rail at number 25, 15 3V3 in bottom + rail at number 25, 16 SDA in B18, 17 SCL in B19. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 18-21, YELLOW: 18 GND in top − rail at number 9, 19 HOLD in I5, 20 5V in top + rail at number 9, 21 LED− Y in J10. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 22-25, RED: 22 GND in top − rail at number 23, 23 DISC in J4, 24 5V in top + rail at number 23, 25 LED− R in J26. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 26-28, BUZZER: 26 GND in top − rail at number 7, 27 3V3 in I3, 28 BUZ in J8. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 29-32, NFC: 29 GND in bottom − rail at number 28, 30 3V3 in bottom + rail at number 28, 31 SDA in E18, 32 SCL in C19. Solder each wire from the top, then tie the bundle to the board edge.
- Leave the + rail link pads beside the left M3 hole open: the top + rail is 5V and the bottom + rail is 3V3. Joining them would short 5V to 3V3. Check with a meter: no short from 5V, 3V3 or SDA/SCL to GND, no short from 5V to 3V3. Then plug U1 in, USB-C toward the board edge as drawn, after checking it is a power-right module. Plug U2 in with its screw terminal toward the bottom edge.

Parts fitted only when needed (marked DNP on the drawing): leave their holes free.

- Part 5, RSD1 (10 kΩ), bottom + rail at number 14 and A14: fit it only if the amplifier's SD pin reads near 0 V in the SD pull-up check.

No.	Item	Holes
	− rail link	right link pad, upper to right link pad, lower
	wire 3V3	G3 to bottom + rail at number 21 (one bend)
	wire LAMP R	E3 to F24 (along the channel)
	wire SCL	E2 to E19 (along the channel)
	wire SDA	C1 to C18
	wire BCLK	B6 to B11
	wire DIN	A8 to A12
	wire LRC	E7 to E10
	wire GND	A15 to bottom − rail at number 15
	wire 5V	top + rail at number 16 to E16
	wire GND	J2 to top − rail at number 2
	wire GND	J12 to top − rail at number 12
	wire GND	J28 to top − rail at number 28
	wire 5V	J1 to top + rail at number 1
1	RG1 220 Ω	F7 to F11
2	RG2 220 Ω	I24 to I27
3	RP1 100 kΩ	J11 to top − rail at number 11
4	RP2 100 kΩ	J27 to top − rail at number 27
5	RSD1 10 kΩ (fit only if needed)	bottom + rail at number 14 and A14
6	Q1 BS170	D H10, G H11, S H12
7	Q2 BS170	D G26, G G27, S G28

No.	Item	Holes
8	U1 sockets	D1-D8, H1-H8
9	U2 socket	D10-D16
10-13	OLED wires	10 GND bottom − rail at number 22, 11 3V3 bottom + rail at number 22, 12 SDA D18, 13 SCL D19
14-17	AIR wires	14 GND bottom − rail at number 25, 15 3V3 bottom + rail at number 25, 16 SDA B18, 17 SCL B19
18-21	YELLOW wires	18 GND top − rail at number 9, 19 HOLD I5, 20 5V top + rail at number 9, 21 LED− Y J10
22-25	RED wires	22 GND top − rail at number 23, 23 DISC J4, 24 5V top + rail at number 23, 25 LED− R J26
26-28	BUZZER wires	26 GND top − rail at number 7, 27 3V3 I3, 28 BUZ J8
29-32	NFC wires	29 GND bottom − rail at number 28, 30 3V3 bottom + rail at number 28, 31 SDA E18, 32 SCL C19